

6. The hormone related to drought tolerance is –

- (a) Absciscic acid (b) Gibberellin
(c) Indole acetic acid (d) Cytokinin

U.P. Lower Sub. (Pre) 2013

Ans. (a)

The hormone absciscic acid was discovered in the early 1960s and plant biologists have known for decades that it plays a crucial role in keeping plants alive during the drought. Absciscic acid causes the temporary closure of stomata when there is a shortage of water. It is very useful for the plants growing in water deficient soil. It functions as antitranspirant and also known as the stress hormone.

7. Which hormone is found in the gaseous form ?

- (a) Absciscic acid (b) Ethylene
(c) Gibberellins (d) Auxins
(e) None of these

Chhattisgarh P.C.S (Pre) 2013

Ans. (b)

Ethylene is a phytohormone (plant hormone) which is found in gaseous form.

8. The gas used in the artificial ripening of green fruit is :

- (a) Acetylene (b) Ethane
(c) Hydrogen (d) Carbon dioxide

U.P.P.C.S. (Pre) 1996

Ans. (a)

Under prevention of food adulteration rule, 1955 (Clause 44AA) acetylene gas can not be used in the ripening of fruits. Calcium carbide is used in some countries as a source of acetylene gas, which is an artificial ripening agent. However, acetylene is not nearly as effective as ethylene and is not a natural plant hormone like ethylene. Also, calcium carbide may contain traces of arsenic and phosphorus, both highly toxic to humans and the use of this chemical for ripening is illegal in most countries. Ethylene has been found not harmful or toxic to humans in the concentrations found in ripening rooms.

9. The gas used for artificial ripening of green fruits is–

- (a) Ethane (b) Acetylene
(c) Carbon-dioxide (d) Oxytocin

Jharkhand P.C.S. (Pre) 2010

U.P. Lower Sub (Pre) 2008

U.P.P.C.S. (Mains) 2007

Ans. (b)

See the explanation of above question.

10. Which of the following phytohormone is mainly responsible for fruit ripening ?

- (a) Cytokinin (b) Absciscic acid
(c) Ethylene (d) None of the above

U.P.P.C.S. (Mains) 2009

U.P.P.C.S. (Pre) 2006

Ans. (c)

See the explanation of above question.

11. It is possible to produce seedless tomato fruits by :

- (a) Applying trace elements in tomato fruits
(b) Spraying mineral solution on plants
(c) Spraying hormones on flowers
(d) Applying fertilizers containing radioactive elements

I.A.S. (Pre) 1996

Ans. (c)

Parthenocarp is also desirable in fruit crops that may be difficult to pollinate or fertilize, such as tomato and summer squash. When sprayed on flowers, any of the plant hormones gibberellin, auxin and cytokinin could stimulate the development of parthenocarpic (seedless) fruit. This is termed artificial parthenocarp.

IV. Plant Reproduction

Notes

- Plant reproduction is the production of new offspring in plants, which can be accomplished by asexual or sexual reproduction. **Asexual reproduction** produces new individuals without the fusion of gametes, genetically identical to the parent plants and each other, except when mutations occur. **Sexual reproduction** produces offspring by the fusion of gametes, resulting in offspring genetically different from the parent or parents.

Asexual Reproduction in Plants :

- Plants have two main types of asexual reproduction in which new plants are produced that are genetically identical clones of the parents. **Vegetative propagation**, which involves a vegetative piece of original plant (budding, tillering, etc.) and **apomixis**, which is a replacement for sexual reproduction. Seeds generated by apomixis are a means of asexual reproduction, involving the formation and dispersal of seeds that do not originate from the fertilization of the embryos.
- Apomixis is the mechanism of seeds production without involving the process of meiosis and syngamy.

- It plays an important role in hybrid seeds production. The method of producing hybrid seeds by cultivation is very expensive for farmers. Also, by sowing hybrid seeds, it is difficult to maintain hybrid characters segregate during meiosis. Apomixis prevents the loss of specific characters in the hybrid. It is cost effective method for producing hybrid seeds.
- In case of apomixis, no meiotic division and fertilization of gametes to form a zygote takes place. The entire process is cut short and seeds formation occurs. It can be of two types–
 - (1) **Sporophytic**–In this type apomixis occurs from the diploid sporophyte.
 - (2) **Gametophytic**–In this type, apomixis occurs from the haploid gametophyte.
- Since the apomixis involves the formation of seeds without syngamy, all the seeds are genetically similar.

Stem Cutting –

- A Plant Cutting is a piece of a plant that is used in horticulture for vegetative propagation. A piece of stem of the plant is placed in a suitable medium such as moist soil. If the conditions are suitable, the plant piece will begin to grow as a new plant, independent of the parent. This technique is adopted in sugarcane.

Vegetative Propagation by Layering–

- The stem that is attached to their parent plant is capable of forming roots on coming in contact with a rooting medium.
- A layer is a rooted stem following detachment from the parent plant. Many plants like strawberry, raspberry and jasmine are propagated by natural layering method.

Grafting –

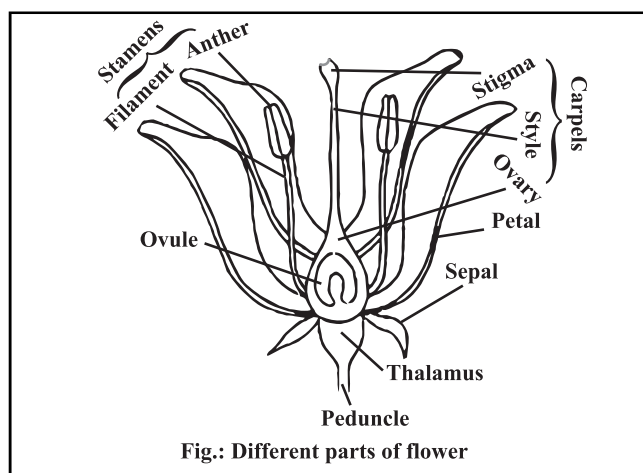
- Grafting is a horticulture technique whereby tissues of plants are joined so as to continue their growth together. The upper part of the combined plant is called the scion while the lower part is called the root-stock.
- The success of this joining requires that the vascular tissue grow together and such joining is called inosculation. This technique is most commonly used in asexual propagation of commercially grown plants for the horticultural and agricultural trades.

Sexual Reproduction in Plants :

- Sexual reproduction in plants involves two fundamental processes : **meiosis**, which rearrange the genes and reduces the number of chromosomes, and **fertilization**, which restores the chromosomes to a complete diploid number.
- Flowers are the sexual organs of the plant and help in sexual reproduction.

Parts of flowers :

- Bisexual flowers typically contain male and female parts in it.
- There are four main layers of the parts of the flower–
- **Calyx** : It is a collection of **sepals** forming the first circle of the flower. These are typically green coloured. In some cases, the sepals have a colour called petaloid. Their main function is to protect the flower while it is still in the bud stage.
- **Corolla** : This layer is the collection of **petals**. It is the second circle of the flower, superior to the calyx. The petals are the colourful part of a flower that helps to attract insects and birds to the flower to facilitate pollination.
- **Androecium** : It is the third layer of flower superior to the corolla. It is the male part of the sexual reproduction of a plant. The androecium is made up of a collection of **stamens**. Each stamen has the following parts–
 - (i) **Anther** : It is present at the tip of the filament. It is internally lobed. Pollen grains are formed inside the anther.
 - (ii) **Filament** : It is a thin stalk-like structure that holds the anther.
- **Gynoecium** : It is the fourth layer of flowers. It is a collection of **carpels** and considered the female reproductive part of the plant. It has three parts–
 - (i) **Stigma** : It is a small and sticky landing structure. The pollen grains from the same or different flower stick to it. This structure acts as a landing for the insects or birds that act as a pollinating agent.
 - (ii) **Style** : It is a thin stalk-like structure that holds the stigma. Two or more carpels may be fused together to varying degrees and the entire structure, including the fused styles and stigmas may be called a **pistil**.
 - (iii) **Ovary** : It is a base of the style or the lower part of the pistil and contains the ovules, which contain the female gamete.



Pollination and Fertilization :

- The transfer of pollen grains from the anther of one flower to the stigma of the same or another flowers is known as pollination. It can be caused by insects, birds, wind, water, and animals including human. These are together called as pollinating agent.

Types of Pollination :

- **Self-Pollination** : Self-pollination is when the pollen of one flower transfers to the stigma of the same flower or other flowers of the same plant. Many flowers that are hermaphrodite (bisexual) see this kind of pollination. Many flowers have various mechanism to prevent self-pollination or promote cross-pollination because the seeds formed as a result of self-pollination are not of genetic variation and healthy.
- **Cross-Pollination** : Cross-pollination is when the pollen of one flower transfers to the stigma of flowers of another plant. This type of pollination brings about genetic variation in the species and allow the plant to withstand changes in the environment better.
- Once the pollen has landed on the stigma of a flower, the pollen tubes develop to transfer the pollen to the ovules which contain the female gamete.

Dichogamy :

- The ripening of stamens and pistils of a flower at different times, so that self-fertilization is prevented, known as dichogamy.
- It is of two types—
 - (a) **Protandry**— Stamens mature before ovary.
 - (b) **Protogyny**— Ovary matures prior to stamens.
- Megasporogenesis results in the formation of female gametes and microsporogenesis results in the formation of male gametes.

Microsporogenesis :

- The anthers contain the pollen mother cell (2n-diploid) that undergoes meiosis to form microspores.
- Tetrad is the result of the microspore mother cell dividing and formation of 4 microspores.
- The anther releases the microspores/pollen grains when it is mature.

Megasporogenesis :

- Megasporangium are the ovules. They are in the ovary and contain the female gametes. Megasporeogenesis is the formation of megaspores from megaspore mother cell (diploid). The resultant of the meiosis of megaspore mother cell is 4 haploid megaspore cells. Of the four cells that form, only one is functional while other degenerate.

Double Fertilization :

- It happens in angiosperms (flowering plants). This is because the male gamete that enters the ovule has two nuclei, one of the male gametes fuses with the female gamete to form a diploid zygote, whereas the other one forms a triploid (3n) endosperm by fusing with the diploid polar nuclei. The zygote divides to form future plant whereas the endosperm provides nutrition to the developing embryo. After fertilization ovary becomes fruits and ovules become the seeds.

Question Bank

1. To study meiosis in plants, the best part would be :

- (a) Shoot apex (b) Root apex
- (c) Anthers (d) Leaf cells

M.P.C.S. (Pre) 2017

Ans. (c)

To study meiosis in plants, the best part would be anthers. In this, the pollination is formed after the meiosis which is actually immature male gametophyte. Stamens are in the form of male genital and carpels are in the form of egg-born female.

2. Which of the following statements is/are correct regarding vegetative propagation of plants?

1. Vegetative propagation produces clonal population.
2. Vegetative propagation helps in eliminating the virus.
3. Vegetative propagation can be practiced most of the year.

Select the correct answer using the code given below.

- (a) 1 only (b) 2 and 3 only
- (c) 1 and 3 only (d) 1, 2 and 3

I.A.S. (Pre) 2014

Ans. (c)

Vegetative or clonal propagation is an asexual reproduction in which successive mitosis of specialized vegetative propagules (as bulbs, corms, tubers, cuttings, buds and apomictic seeds) develop new plants and results in a clonal population. Therefore statement 1 is correct. Viruses are transmitted from plant to plant in a number of ways including vegetative propagation. Viral diseases could be transferred through vegetative parts. So the statement 2 is incorrect. Vegetative propagation can be practised most of the year. Hence, statement 3 is correct.

3. The practice of gootee in plants is used for the purpose of :

- (a) Control of insects
- (b) For vegetative propagation

- (c) Germination of seeds
(d) Weed control

U.P. R.O./A.R.O. (Pre) 2021

Ans. (b)

Gootee in plants is a method of vegetative propagation of plants that requires the rooting of aerial stems when connected to the parent plant. It is also named as air layering, marcotting, pot layerage, marcottage and circumposition.

4. With reference to the current trends in the cultivation of sugarcane in India, consider the following statements:

1. A substantial saving in seed material is made when 'bud chip settlings' are raised in a nursery and transplanted in the main field.
2. When direct planting of setts is done, the germination percentage is better with single-budded setts as compared to setts with many buds.
3. If bad weather conditions prevail when setts are directly planted, single-budded setts have better survival as compared to large setts.
4. Sugarcane can be cultivated using settlings prepared from tissue culture.

Which of the statements given above is/are correct?

- (a) 1 and 2 only (b) 3 only
(c) 1 and 4 only (d) 2, 3 and 4 only

I.A.S. (Pre) 2020

Ans. (c)

With reference to the current trends in the cultivation of sugarcane in India:

- ♦ The 'bud chip' method for cane plantation reduces the mass and enables quick multiplication of seeds. This method has proved to be more economical and convenient than the traditional way of planting two or three bud sets. The returns are relatively better and a substantial saving in seed material is made when 'bud chip settlings' are raised in a nursery and transplanted in the main field.
- ♦ When direct planting of setts is done, it is found that the germination percentage is better with many-buds setts as compared to single-budded setts.
- ♦ When setts are directly planted, large setts have better survival under bad weather than single-budded setts.
- ♦ Planting material for vegetative propagation of sugarcane includes whole cane, setts having of single buds to many buds, setts prepared in nursery or setts prepared from tissue culture.

Hence, among the given statements, only statements 1 and 4 are correct.

5. Sugarcane is a :

- (a) short day plant (b) long day plant
(c) day neutral plant (d) photo insensitive plant

Jharkhand P.C.S. (Pre) 2023

Ans. (a)

Sugarcane plants usually bloom in winter, when days are shorter. It is therefore a short day plant as classified by photoperiodism. In general, sugarcane is termed an intermediate daylength plant with quantitative short-day behavior.

6. With reference to crop rotation, which of the following statement is/are correct?

1. Deep rooted crops should be followed by the same type of crop.
2. Legume must be followed by a non-legume.

Select the correct answer using the codes given below :

Code :

- (a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

U.P. R.O./A.R.O. (Pre) 2021

Ans. (b)

Crop rotation refers to the cultivation of different crops successively, in line with changing seasons. The order and area of crops to be grown is accordingly patterned for optimizing soil nutrients and water, controlling pests and suppressing weeds. An effective crop rotation system requires right selection of crops that adjusts well with the space, sunlight and soil in rooftop garden. General principles that guide crop rotation are listed hereafter :

- ♦ Leguminous crops should be followed by non-leguminous crops.
- ♦ The crops with tap roots (dicots) should be grown after those which have fibrous root system (monocots).
- ♦ Slow-growing crops which are more vulnerable to weeds should be grown immediately after weed-suppressing crops in a rotation system.
- ♦ Closely related species should not be grown successively as they act as alternate hosts for insects, pests and diseases.
- ♦ Deep rooted crops should be succeeded by shallow rooted crops.
- ♦ Crops susceptible to soil borne pathogens and parasitic weeds should be followed by tolerant crops.

7. Given below are two statements, in which one is labelled as Assertion (A) and the other as Reason (R).

Assertion (A) : Pulses, when included in intercropping with cereals, act as complimentary crops.

Reason (R) : Pulses with tap root system, do not compete for nutrients and moisture with cereals having fibrous root system.

Select the correct answer from the code given below :

Code :

- (a) (A) is true but (R) is false.
- (b) (A) is false but (R) is true.
- (c) Both (A) and (R) are true but (R) is not correct explanation of (A).
- (d) Both (A) and (R) are true and (R) is correct explanation of (A).

U.P. R.O./A.R.O. (Pre) 2023

Ans. (c)

Intercropping is the practice of growing two or more crops in proximity. Pulses are generally intercropped with coarse cereals and oilseeds under rainfed condition in our country. Pulses, when included in intercropping with cereals, act as complimentary crops. The slow initial growth habit and deep tap root system makes these crops more suitable for intercropping with cereals and oilseed crops than any other crop. Pulses with tap root system, do not compete for nutrients and moisture with cereals having fibrous root system. The tap root system allows plants to anchor better, get sources of minerals and water from further into the ground than fibrous-rooted plants. Apart from increasing profitability and resource use efficiency, pulses as intercrop act as safeguard under unprecedented moisture stress condition. Pulses can contribute significantly to achieve the twin objectives of increasing productivity and improving the sustainability of rice and wheat based cropping systems. The pulses grown together with or before a cereal crop can reduce and sometimes substitute the need of nitrogen application as well. Hence, both (A) and (R) are true but (R) is not the correct explanation of (A).

8. Clonal selection can be used in –

- (a) Groundnut
- (b) Mustard
- (c) Wheat
- (d) Potato

U.P.P.C.S. (Mains) 2009

Ans. (d)

The procedure of selection used for asexually propagated crops is known as clonal selection since the selected plants are used to produce new clones. This method is used in the production of potatoes.

9. A plant bud is –

- (a) An embryonic shoot
- (b) An embryonic leaf
- (c) An endosperm
- (d) A seed

56th to 59th B.P.S.C. (Pre) 2015

Ans. (a)

In botany, a bud is an undeveloped or embryonic shoot and normally occurs in the axil of a leaf or at the tip of the stem.

10. Embryo is found in –

- (a) Flowers
- (b) Leaves
- (c) Seeds
- (d) Buds

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (c)

The embryo along with other cells from the mother plant develops into the seed for the next generation, which, after germination, grows into a new plant.

11. Match List-I with List-II and select the correct answer by using the codes given below the lists :

List-I	List-II
A. Fruit	1. Ovule
B. Seed	2. Leaf
C. Wood	3. Stem
D. Starch	4. Ovary

Code :

	A	B	C	D
(a)	2	1	3	4
(b)	4	1	3	2
(c)	2	3	1	4
(d)	4	3	1	2

I.A.S. (Pre) 1998

Ans. (b)

The correctly matched lists are as follows :

Fruit	–	Ovary
Seed	–	Ovule
Wood	–	Stem
Starch	–	Leaf

12. Plants are formed like parent plant from –

- (a) Seeds
- (b) Stem cut
- (c) None of these
- (d) Both of these

39th B.P.S.C. (Pre) 1994

Ans. (b)

Plants are propagated by two methods –

1. Sexual propagation or seed propagation.
2. Asexual or vegetative propagation.

Vegetative propagation is a form of asexual reproduction of a plant. Only one plant is involved and offspring is the result of one parent. The new plant is genetically identical to the parent. In this process, new organism arises from any vegetative part of the parent (root, stem, leaf etc.) and possesses exactly the same characteristics of the parent plant.

The purity of plants is maintained in vegetative propagation which is not possible in sexual reproduction. Stem cutting, grafting, layering, etc. are common artificial methods of vegetative propagation.

13. Stem cutting is done generally to grow –

- (a) Banana
- (b) Sugarcane
- (c) Mango
- (d) Cotton

39th B.P.S.C. (Pre) 1994

Ans. (b)

Botanically, Sugarcane belongs to the Andropogoneae tribe of the family Gramineae. Sugarcane plants are propagated by stem cutting. Stem cutting is segmented parts of the stem or entire stems detached from the main branch. Mango and cotton are generally grown from seeds while suckers are used as the planting material for banana. Sucker in banana arises from the root and is an extension of the rhizome as a lateral root. Some farmers also practice tissue culture for banana cultivation using seedlings developed through tissue culture. Rhizomes and peepers are the other planting materials used for banana cultivation.

14. Consider the following plants :

- 1. Bougainvillea
- 2. Carnation
- 3. Cocoa
- 4. Grapes

Which of these plants are propagated by stem cuttings ?

- (a) 1 and 2
- (b) 2, 3 and 4
- (c) 1, 2 and 4
- (d) 1, 2, 3 and 4

I.A.S. (Pre) 2002

Ans. (d)

Propagation of plants by stem cuttings is the most popular plant propagation method for woody shrubs and ornamental plants. This is also a good technique for house plants. Some of the plants propagated by stem cuttings are Sugarcane, grapes, Rose, Cocoa, Carnation, Duranta, Bougainvillea, etc. Although in plants like Cocoa, propagation by seeds is also popular.

15. Consider the following statements:

- 1. Sweet orange plant is propagated by grafting technique.

- 2. Jasmine plant is propagated by layering technique.

Which of the statement(s) given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

I.A.S. (Pre) 2009

Ans. (c)

Layering is a way to grow new plants as in case of Jasmine. (from existing plants without having to take any cuttings). In a nutshell, bury part of a stem or branch in the soil and new roots and shoots will form. Grafting or graftage is a horticultural technique whereby tissues from one plant are inserted into those of another so that the two sets of vascular tissues may join together. Sweet orange plant is propagated by this technique.

16. Buds are typically found :

- (a) Both at the base of leaves and tip of branches
- (b) At the tip of branches
- (c) At the base of leaves
- (d) None of the above

70th B.P.S.C. (Pre) (Re-Exam) 2024

Ans. (a)

A plant bud is an undeveloped or embryonic shoot. Buds are typically found in the axil of a leaf (base of the leaf where leaf joins the stem) or at the tip of a stem or branch. These are known as axillary or terminal buds, respectively. Axillary buds are located at the angle between a leaf and the stem. They can develop into branches, flowers, or new stems. Terminal buds are located at the tip of a stem or branch and are responsible for the vertical growth of the plant. While less common, adventitious buds can also arise in unexpected places like on roots or leaves. These are referred to as radical buds (on roots) or epiphyllous buds (on leaves), respectively.

17. The most important part of the plant in term of its life cycle is–

- (a) Flower
- (b) Leaf
- (c) Stem
- (d) Root

47th B.P.S.C. (Pre) 2003

Ans. (a)

The flower is the most important part of the plant in term of its life cycle. The biological function of a flower is to mediate the union of male and female gametes in order to produce seeds. The process begins with pollination followed by fertilization and continues with the formation and dispersal of the seed.

18. Which of the three circumstances among these are most necessary in the germination of single seed plants?

- (a) Soil, water, oxygen
- (b) Water, proper temperature, oxygen
- (c) Proper temperature, oxygen, CO₂
- (d) Temperature, oxygen, light.

Uttarakhand U.D.A./L.D.A. (Pre) 2003

Ans. (b)

Water, proper temperature and oxygen are the most required substances/circumstances for the germination of seed, while the sunlight is not essential. There are two types of germination of seed in plants - epigeal and hypogeal.

19. Which one is not required for the germination of seed?

- (a) Light
- (b) Humidity
- (c) Proper temperature
- (d) Oxygen

U.P.P.C.S. (Pre) 1990

Ans. (a)

See the explanation of above question.

20. Censer mechanism of dispersal of seeds is found in :

- (a) Pea
- (b) Poppy
- (c) Cotton
- (d) Maize

U.P.P.C.S. (Mains) 2010

Ans. (b)

Seed dispersal is the movement or transport of seeds away from the parent plant. Plants have very limited mobility and consequently rely upon a variety of dispersal vectors to transport their propagules, including both abiotic and biotic vectors. Censer mechanism of dispersal of seeds is found in poppy.

21. Transfer of pollen grains from the anther to stigma of another flower of the same plant is called

- (a) Autogamy
- (b) Geitonogamy
- (c) Xenogamy
- (d) None of the above

M.P.P.C.S. (Pre) 2019

Ans. (b)

Transfer of pollen grains from the anther to stigma of another flower of the same plant is called geitonogamy. It is a type of self-pollination.

22. Consider the following kinds organisms –

- 1. Bat
- 2. Bee
- 3. Bird

Which of the above is/are pollinating agent/agents?

- (a) 1 and 2
- (b) Only 2
- (c) 1 and 3
- (d) 1, 2 and 3

I.A.S. (Pre) 2012

Ans. (d)

Pollination is the act of transferring pollen grains from male anther of a flower to the female stigma. It is a prerequisite for fertilization. Fertilization allows the flower to develop fruits and seeds. Bees are the most important pollinating agent. Pollination by bees and butterflies often occurs in the flowers which bloom in the day time. Hummingbird-pollinated flowers appear in warm colours like red, yellow and orange. Bat is also a pollinating agent of many flowers. However the number of animals - pollinated flowers are not so high, some of them are found in the southern hemisphere.

23. Pollination is best defined as :

- (a) transfer of pollen grain from anther to stigma
- (b) germination of pollen grains
- (c) growth of pollen tube in ovule
- (d) visiting flowers by insects
- (e) None of the above/More than one of the above

67th B.P.S.C. (Pre) (Re. Exam) 2022

Ans. (a)

See the explanation of above question.

24. Seed that is changed every year is known as -

- (a) Breeder seed
- (b) Certified seed
- (c) Foundation seed
- (d) Hybrid seed

U.P.P.C.S. (Mains) 2010

Ans. (d)

Hybrid seed is changed every year. In agriculture and gardening, hybrid seed is seed produced by cross pollinated plants. Hybrid seed production is predominant in agriculture and home gardening.

25. Consider the following events of seed multiplication stages and arrange these in correct chronological order from earliest stage to last.

- I. Breeder seed
- II. Foundation seed
- III. Certified seed
- IV. Nucleus seed

Select the correct answer from the code given below :

Code :

- (a) III, I, IV, II
- (b) II, III, I, IV
- (c) II, IV, III, I
- (d) IV, I, II, III

U.P. R.O./A.R.O. (Pre) 2023

Ans. (d)

The correct chronological order of events of seed multiplication stages from earliest stage to last, is as follows :
Nucleus seed → Breeder seed → Foundation seed → Certified seed.
Hence, option (d) is the correct answer.